

CH:3 PARAMETER ESTIMATION METHODS

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What is Parameter Estimation?

Parameter estimation is a process used in statistics and machine learning to determine the **parameters of a probability distribution or statistical model** from a given dataset.

- In pattern recognition, models often assume the data comes from a known distribution (e.g., Gaussian), but the **parameters (like mean, variance)** of that distribution are **unknown**.
- Parameter estimation aims to **infer these parameters** using observed data samples.

In formal terms:

Given a dataset $\mathbf{X} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$ assumed to be drawn from a distribution $\mathbf{P}(\mathbf{x} | \boldsymbol{\theta})$ with unknown parameters $\boldsymbol{\theta}$, parameter estimation seeks to determine $\hat{\boldsymbol{\theta}}$, an estimate of $\boldsymbol{\theta}$.

Importance in Statistical Modeling

1. **Model Fitting:** Fits theoretical distributions to real-world data.
2. **Prediction:** Enables prediction using the fitted model.
3. **Classification and Clustering:** Required in probabilistic models like Bayesian classifiers and GMM.
4. **Performance Optimization:** Helps models perform optimally by tuning parameters.
5. **Learning from Data:** Forms the foundation of most machine learning methods.

Types of Parameter Estimation

1. **Point Estimation:** A **single value** is computed as the best estimate of an unknown parameter.

Examples:

- Sample mean as an estimate of population mean
- Maximum likelihood estimation (MLE)
- Method of moments

Desirable properties of point estimators:

- **Unbiasedness:** Expected value of estimator equals the true parameter.
- **Consistency:** Estimator converges to the true parameter as sample size increases.
- **Efficiency:** Estimator has the minimum possible variance among all unbiased estimators.

2. **Interval Estimation:** Provides a **range of values** (an interval) within which the parameter is likely to lie, with a specified confidence level. Example:

- 95% confidence interval for a mean: $\mu \in [\bar{x} - 1.96\sigma/\sqrt{n}, \bar{x} + 1.96\sigma/\sqrt{n}]$

Key terms:

- **Confidence Interval (CI):** The range computed from the sample that, with a certain probability, contains the true parameter.
- **Confidence Level:** Probability that the CI contains the true parameter (e.g., 95%).

Applications in Pattern Recognition

1. **Supervised Learning:** Used in estimating priors and likelihoods for classifiers.
2. **Unsupervised Learning:** GMMs use it to estimate component parameters.
3. **Feature Extraction:** PCA uses estimated covariance for dimensionality reduction.
4. **Density Estimation:** Needed in both parametric and non-parametric methods.
5. **Speech/Image Recognition:** HMMs and GMMs require parameter estimation for modeling sequences and features.

Maximum-Likelihood Estimation (MLE)

Concept of Likelihood Function, Log likelihood function and MLE

Objective: To estimate the **parameters (θ)** of a statistical model that make the observed data **most probable**.

Given a set of **independent and identically distributed (i.i.d.)** data samples $\mathbf{X} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$ and a probability density function (pdf) or probability mass function (pmf) $\mathbf{P}(\mathbf{x} | \theta)$ parameterized by θ , the **likelihood function** is defined as:

$$\mathbf{L}(\theta | \mathbf{X}) = \prod_i \mathbf{P}(\mathbf{x}_i | \theta)$$

This function is treated as a function of θ , with the data $\mathbf{X}$ fixed.

Log-Likelihood: To simplify computation, we usually take the **logarithm** of the likelihood function (log-likelihood). This converts products into sums and is easier to differentiate:

$$\log L(\theta | X) = \sum_i \log P(x_i | \theta)$$

Maximum Likelihood Estimate (MLE): The MLE of θ is the value $\hat{\theta}$ that **maximizes the likelihood function**, or equivalently, the log-likelihood:

$$\hat{\theta} = \operatorname{argmax}_{\theta} \log L(\theta | X)$$

This is typically done by taking the derivative of the log-likelihood with respect to θ , setting it to zero, and solving for θ .

Properties of MLE

1. **Consistency:** As sample size increases ($n \rightarrow \infty$), the MLE converges to the true parameter value.
2. **Asymptotic Normality:** For large n , the distribution of the MLE approaches a normal distribution centered at the true parameter.
3. **Efficiency:** Among all unbiased estimators, MLE has the **lowest variance** in the limit (it reaches the Cramér-Rao Lower Bound under regularity conditions).
4. **Invariance:** If $\hat{\theta}$ is the MLE of θ , and $g(\theta)$ is a function of θ , then $g(\hat{\theta})$ is the MLE of $g(\theta)$.
5. **Not Always Unbiased:** MLEs can be **biased**, especially for small samples, though they become **asymptotically unbiased**.
6. **Computational Simplicity:** Often leads to closed-form solutions (e.g., sample mean for Gaussian), but can be complex in some models (requiring iterative methods like EM).

Finding Parameters by Maximizing Likelihood

Find the parameter(s) θ of a probability distribution that **maximize the likelihood** of the observed data. **Steps in Finding MLE:**

1. **Write the likelihood function** based on the assumed distribution.
2. **Take the natural logarithm** of the likelihood to get the **log-likelihood**.
3. **Differentiate the log-likelihood** with respect to the parameter(s).
4. **Set derivative(s) to zero** and solve for the parameter(s).
5. **Verify maximum** (using second derivative or logic).

Derivation of MLE for Mean & Variance for Gaussian Distribution

① Assuming

→ Dataset $X = [x_1, x_2, \dots, x_n]$

→ Distribution: $x_i \sim \mathcal{N}(\mu, \sigma^2)$

→ Parameters to estimate: μ and σ^2

Step 1: Write the Likelihood func.

$$L(\mu, \sigma^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x_i - \mu)^2}{2\sigma^2}}$$

Step 2: Taking log-likelihood

$$\begin{aligned} \log(L(\mu, \sigma^2)) &= -\log\left(\prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}}\right) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2 \\ &= -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2 \end{aligned}$$

Step 3: We find Partial Derivatives & equate to 0.

For mean

$$\begin{aligned} \frac{\partial}{\partial \mu} (\log L) &= -\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu) (-2) \\ &= \frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu) = \frac{1}{\sigma^2} \left[\sum_{i=1}^n x_i - n\mu \right] \end{aligned}$$

$$\therefore \frac{1}{\sigma^2} \left[\sum_{i=1}^n x_i - n\mu \right] = 0$$

$$\Rightarrow \boxed{\mu = \frac{1}{n} \sum_{i=1}^n x_i}$$

For variance

$$\frac{\partial (\log L)}{\partial \sigma^2} = -\frac{n}{2} \frac{1}{\sigma^2} - \frac{1}{2} \frac{\partial}{\partial \sigma^2} (\sigma^{-2}) \sum_{i=1}^n (x_i - \mu)^2$$

$$= -\frac{n}{2\sigma^2} - \frac{1}{2} (-2) \sigma^{-4} \sum_{i=1}^n (x_i - \mu)^2$$

$$= -\frac{n}{2\sigma^2} + \frac{1}{\sigma^4} \sum_{i=1}^n (x_i - \mu)^2 = 0$$

$$\Rightarrow \frac{1}{\cancel{2\sigma^4}} \sum_{i=1}^n (x_i - \mu)^2 = \cancel{2\sigma^2}$$

$$\Rightarrow \boxed{\sigma^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2}$$

w.r.t covariance

$$\frac{\partial}{\partial \Sigma} (a^T \Sigma^{-1} a) = -\Sigma^{-1} a a^T \Sigma^{-1} \quad (\text{rule})$$

$$\therefore \frac{\partial (\log \mathcal{L})}{\partial \Sigma} = -\frac{n}{2 \Sigma} + \frac{1}{2} \Sigma^{-1} \left(\sum_{i=1}^n (x_i - \mu)(x_i - \mu)^T \right) \Sigma^{-1}$$

$$\Rightarrow \frac{n}{2 \Sigma} = \frac{1}{2} \Sigma^{-1} \left(\sum_{i=1}^n (x_i - \mu)(x_i - \mu)^T \right) \Sigma^{-1}$$

$$\Rightarrow n = \Sigma^{-1} \left(\sum_{i=1}^n (x_i - \mu)(x_i - \mu)^T \right) \Sigma^{-1}$$

$$\Rightarrow \Sigma = \frac{1}{n} \sum_{i=1}^n (x_i - \mu)(x_i - \mu)^T$$

Gaussian Mixture Models (GMM)

A **Gaussian Mixture Model (GMM)** is a **probabilistic model** that assumes that the data is generated from a mixture of **several Gaussian distributions** (also called components), each with its own parameters.

Each component in the mixture represents a **sub-population** or **cluster**, and the overall data distribution is a **weighted sum** of these individual Gaussians.

Use in Clustering:

- GMM is used in **unsupervised learning** for clustering data based on the assumption that each cluster is generated by a different Gaussian distribution.
- Unlike k-means, GMM provides **soft clustering**, meaning it gives a **probability** of each point belonging to each cluster.
- It is more **flexible** than k-means since it allows **elliptical clusters** (via covariance), not just spherical ones.

Parameters in GMM

Each component Gaussian k has three main parameters:

a) Mean vector μ_k

- Center of the k^{th} Gaussian component.
- Controls the **location** of the component in feature space.

b) Covariance matrix Σ_k

- Defines the **shape, orientation, and spread** of the Gaussian.
- In 1D, it is just the variance. In 2D or higher, it can model ellipses or general shapes.
- Determines whether the component is **spherical, elliptical, or rotated**.

c) Mixing coefficient π_k

- Represents the **prior probability** of choosing component k .
- Must satisfy $\sum \pi_k = 1$ ($k=1$ to K)
- Indicates the **relative size or weight** of each Gaussian in the mixture.

Probability Density Function of GMM.

$$p(x) = \sum_{k=1}^K \pi_k \mathcal{N}(\mu_k, \Sigma_k)$$

here,

$$\rightarrow \sum_{k=1}^K \pi_k = 1, \pi_k \geq 0$$

$\rightarrow \mathcal{N}(\mu_k, \Sigma_k)$ is multivariate Gaussian pdf.

$$\mathcal{N}(\mu_k, \Sigma_k) = \frac{1}{\sqrt{(2\pi)^d |\Sigma_k|}} e^{-\frac{1}{2} (x - \mu)^T \Sigma^{-1} (x - \mu)}$$

① μ_k = mean vector

② Σ_k = covariance matrix

③ π_k = mixing coefficient
(weight of component k)

How GMM Works in Practice (Summary)

1. A data point x is assumed to be generated by first picking a component k with probability π_k , then sampling x from $\mathcal{N}(\mu_k, \Sigma_k)$
2. The overall distribution is a combination of these individual Gaussians.
3. Algorithms like **Expectation-Maximization (EM)** are used to estimate the parameters, μ_k , Σ_k , π_k from the data.

Applications of GMM

1. Clustering (Unsupervised Learning)

- Groups data into clusters when the number and shape of clusters are unknown.
- More flexible than K-means due to elliptical cluster modeling (via covariance matrices).

2. Speech Recognition

- Models the probability distribution of acoustic features (e.g., MFCCs) for each phoneme.
- Often combined with Hidden Markov Models (HMM-GMM) for sequence modeling in older speech systems.

3. Image Segmentation

- Separates an image into regions based on pixel intensities or colors.
- Each region is modeled as a Gaussian component in RGB or grayscale space.

4. Anomaly Detection

- Fits GMM to "normal" data; points with low likelihood under the model are flagged as anomalies.
- Useful in network security, fraud detection, manufacturing defect identification, etc.

5. Background Subtraction in Videos

- Models the distribution of pixel intensities over time to separate foreground objects from the background.
- Each pixel location is treated as a mixture of Gaussians to handle changes like lighting, movement, etc.

6. Density Estimation

- GMM serves as a smooth, continuous approximation of the underlying probability distribution of data.
- Useful in generative modeling, simulations, and probabilistic reasoning.

7. Bioinformatics

- Clustering gene expression data, modeling population genetics, identifying subtypes in biomedical data.

Expectation-Maximization (EM) algorithm

The **Expectation-Maximization (EM)** algorithm is an **iterative optimization technique** used to find **maximum likelihood estimates** of parameters in statistical models, particularly when the model involves **latent (hidden or unobserved) variables**.

It alternates between:

- **E-step:** Estimate the hidden variables using current parameters.
- **M-step:** Update parameters to maximize the likelihood based on those estimates.

Need for EM in Latent Variable Models

What is a Latent Variable Model?

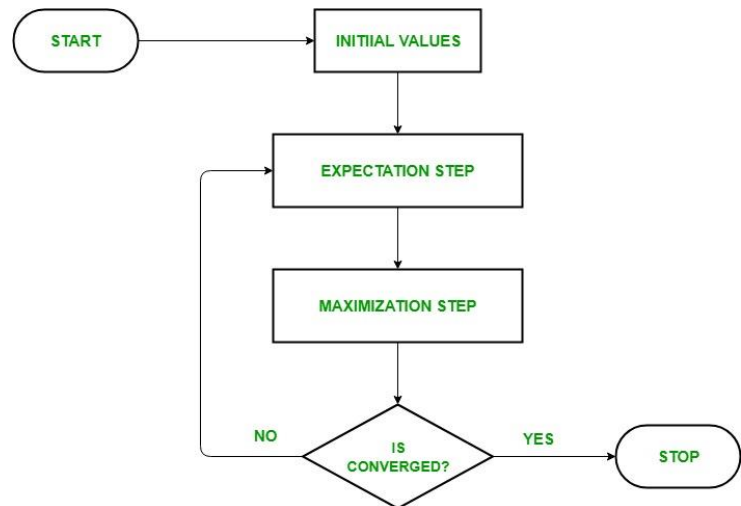
- A model where some variables (**latent variables**) are not directly observed in the data but are assumed to exist.
- In **GMM**, the latent variable is the **cluster/component assignment** (i.e., which Gaussian each data point came from).

Why EM is Needed

- The **log-likelihood** of such models becomes complex because we need to marginalize over the hidden variables.
- **Direct maximization** of the log-likelihood becomes **intractable** due to summation inside the log function.
- EM provides a **tractable iterative solution** to find the **maximum likelihood estimates** of parameters.

EM algorithm in detail.

1. Initially, a set of initial values of the parameters are considered. A set of incomplete observed data is given to the system with the assumption that the observed data comes from a specific model.
2. Expectation step (E-step): Using the observed available data of the dataset, estimate (guess) the values of the missing data. It is basically used to update the variables.
3. Maximization step (M-step): Complete data generated after the expectation (E) step is used in order to update the parameters. It is basically used to update the hypothesis.
4. Repeat step 2 and step 3 until convergence.



Steps in EM Algorithm

Assume

- $X = \{x_1, x_2, \dots, x_n\}$ be observed data
- $Z =$ Latent variables
- θ : parameters of model (mean, covariance, mixing coefficients)

Step 1: E-step

Computing posterior probability that a datapoint x_i was generated by component k ,

$$P(z \in k) = \frac{\pi_k \mathcal{N}(x_i | \mu_k, \Sigma_k)}{\sum_{j=1}^K \pi_j \mathcal{N}(x_i | \mu_j, \Sigma_j)}$$

Step 2: M step

we update the parameters to maximize the expected log likelihood

(i) Mean
$$\mu_k^{(t+1)} = \frac{\sum_{i=1}^n P(x \in k) \cdot x_i}{\sum_{i=1}^n P(x \in k)}$$

(ii) Covariance
$$\Sigma_k^{(t+1)} = \frac{\sum_{i=1}^n P(x \in k) (x_i - \mu_k) (x_i - \mu_k)^T}{\sum_{i=1}^n P(x \in k)}$$

(iii) Mixing Coefficient
$$\pi_k^{(t+1)} = \frac{1}{n} \sum_{i=1}^n P(x \in k)$$

Step 3: Repeat step 1 & 2 until parameters converge or change becomes minimal.

Use in Training Gaussian Mixture Models (GMM)

In GMM:

- We do **not know** which Gaussian component each data point came from → the **latent variable**.
- EM estimates:
 - **Responsibilities** (E-step): Probabilities of assignment to Gaussians
 - **Parameters** (M-step): Means, covariances, and mixing weights of the Gaussians

EM allows us to:

- Fit a GMM to data **even without knowing component labels**
- Perform **soft clustering**
- Estimate full parameters of the mixture model from **unlabelled data**

Uses of EM Algorithm

1. **Gaussian Mixture Models (GMM):** EM is the standard method to estimate parameters in GMM. Handles unknown cluster assignments (latent variables).
2. **Missing Data Problems:** EM estimates parameters when parts of the data are missing or unobserved.
3. **Hidden Markov Models (HMM):** The Baum-Welch algorithm, used to train HMMs, is a type of EM algorithm.

4. **Image Processing & Segmentation:** EM is used to separate image regions probabilistically based on pixel values.
5. **Medical Diagnosis & Bioinformatics:** Used in probabilistic inference when some patient attributes or biological factors are hidden.

Advantages of EM Algorithm

1. **Handles Incomplete Data:** Works well when some variables are unobserved or latent.
2. **Simple to Implement:** Each step (E and M) usually has a straightforward closed-form or numerical solution.
3. **Guaranteed Convergence:** EM guarantees that the likelihood does not decrease with each iteration.
4. **Soft Assignment:** Unlike hard clustering (like k-means), EM gives probabilistic (soft) assignments.
5. **Flexible Models:** EM can be applied to many models beyond GMMs (e.g., HMMs, factor analysis, etc.).

Disadvantages of EM Algorithm

1. **Converges to Local Maxima:** EM can get stuck in local maxima of the likelihood function; results depend on initialization.
2. **Slow Convergence:** May require many iterations to converge, especially near the maximum.
3. **No Guarantee of Global Optimum:** It does not necessarily find the best global parameter estimates.
4. **Sensitive to Initialization:** Poor initial parameter values may lead to poor final results.
5. **Assumes Model Correctness:** Works best when the data truly comes from the assumed model (e.g., mixture of Gaussians). If not, performance degrades.

Bayesian Estimation

Bayesian estimation is a statistical approach that **updates our belief about a parameter** by combining **prior knowledge** with **evidence from observed data** using **Bayes' theorem**.

Unlike frequentist methods (like MLE), Bayesian estimation treats model parameters as **random variables** with **distributions**, not fixed unknown values.

Prior and Posterior Probability

Prior Probability:

- Denoted as $P(\theta)$, it represents our **initial belief** about the parameter θ **before** observing any data.
- Can be based on past experience, domain knowledge, or chosen to be non-informative.

Posterior Probability:

- Denoted as $P(\theta|X)$, it is the **updated belief** about the parameter θ **after** observing data X .
- Computed using **Bayes' theorem**.

Bayes' Theorem in Parameter Estimation

$$P(\theta|X) = \frac{P(X|\theta) \cdot P(\theta)}{P(X)}$$

- $P(\theta|X)$: Posterior probability of parameter given data (what we want)
- $P(X|\theta)$: Likelihood of observing data given the parameter (as in MLE)
- $P(\theta)$: Prior probability (belief before data)
- $P(X)$: Marginal likelihood (normalizing constant), often written as: $P(X) = \int P(X|\theta) \cdot P(\theta) d\theta$

So, Bayesian estimation combines prior belief with observed evidence.

Bayesian vs MLE: When to Prefer Which

Aspect	MLE	Bayesian / MAP
Data Dependency	Uses only observed data	Uses both data and prior
Parameter Assumption	Assumes parameters are fixed	Parameters are random variables
Uncertainty Modeling	No uncertainty; point estimate	Models uncertainty with a distribution
When Preferred	Large datasets; no prior info	Small datasets; prior knowledge is available
Overfitting	More prone to overfitting	Less prone (due to prior regularization)
Interpretability	Easier to interpret and compute	May involve complex integrations

Maximum Likelihood Estimation (MLE):

- Finds the parameter θ that maximizes the likelihood: $\theta_{MLE} = \arg \max_{\theta} P(X|\theta)$
- Does not consider prior knowledge; relies only on data.

Maximum A Posteriori Estimation (MAP):

- Finds the parameter θ that maximizes the posterior: $\theta_{MAP} = \arg \max_{\theta} P(\theta|X) = \arg \max_{\theta} [P(X|\theta) \cdot P(\theta)]$
- Incorporates prior belief about θ .
- If the prior $P(\theta)$ is uniform (flat), then MAP becomes equivalent to MLE.